



Running blades are made of several layers of **carbon fiber**—each layer thinner than a human hair.

PARALYMPIC SPORT

RUNNING BLADES

Artificial lower limbs worn by amputee athletes are called running blades. Rather than copying biological legs, they are designed to be light, curved, and elastic, which means that they return to their original size and shape after being deformed by a force. They behave like leg muscles and tendons—under the wearer's weight, they bend and store energy, then release it as they push off the ground.



EXTRA BOUNCE

The bouncing boost of a trampoline comes from the springs that attach the fabric “bed” to the rigid frame. When a person jumps, their weight forces the springs to extend downward. The springs exert an equal and opposite (upward) force on the person as they return to their original position—sending the person bouncing up into the air.